$$\mathrm{N_2O_4(g) \rightleftharpoons 2\,NO_2(g)}$$

$$\mathrm{2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)}$$

$$\mathrm{H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)}$$

Nuclear Equations

In nuclear notation the mass number is a superscript and the atomic number a subscript, both before the symbol. The nuclide ${}^{14}_{6}\mathrm{C}$ has 6 protons and 8 neutrons. The nuclide ${}^{238}_{92}\mathrm{U}$ has 92 protons and 146 neutrons. The nuclide ${}^{60}_{27}\mathrm{Co}$ has 27 protons and 33 neutrons. The nuclide ${}^{131}_{53}\mathrm{I}$ has 53 protons and 78 neutrons. The nuclide ${}^{56}_{26}\mathrm{Fe}$ has 26 protons and 30 neutrons. The nuclide ${}^{16}_{8}\mathrm{O}$ has 8 protons and 8 neutrons. Alpha decay of U-238:

$$\mathrm{{}^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He}$$

Alpha decay of Ra-226:

$$\mathrm{{}^{226}_{88}Ra \longrightarrow {}^{222}_{86}Rn + {}^{4}_{2}He}$$

Alpha decay of Po-210:

$$\mathrm{{}^{210}_{84}Po \longrightarrow {}^{206}_{82}Pb + {}^{4}_{2}He}$$

Alpha decay of Pu-239:

$$\mathrm{{}^{239}_{94}Pu \longrightarrow {}^{235}_{92}U + {}^{4}_{2}He}$$

Alpha decay of Rn-222:

$$\mathrm{{}^{222}_{86}Rn \longrightarrow {}^{218}_{84}Po + {}^{4}_{2}He}$$

Alpha decay of Th-232:

$$\mathrm{{}^{232}_{90}Th \longrightarrow {}^{228}_{88}Ra + {}^{4}_{2}He}$$

Alpha decay of Am-241:

$$\mathrm{{}^{241}_{95}Am \longrightarrow {}^{237}_{93}Np + {}^{4}_{2}He}$$

Beta decay of C-14:

$$\mathrm{{}^{14}_{6}C \longrightarrow {}^{14}_{7}N + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of H-3:

$$\mathrm{{}^{3}_{1}H \longrightarrow {}^{3}_{2}He + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of Co-60:

$$\mathrm{{}^{60}_{27}Co \longrightarrow {}^{60}_{28}Ni + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of Sr-90:

$$\mathrm{{}^{90}_{38}Sr \longrightarrow {}^{90}_{39}Y + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of I-131:

$$\mathrm{{}^{131}_{53}I \longrightarrow {}^{131}_{54}Xe + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of Cs-137:

$$\mathrm{{}^{137}_{55}Cs \longrightarrow {}^{137}_{56}Ba + {}^{0}_{-1}e + \bar{\nu}_e}$$